T0-invariant form, Curve25519 miniature

Semaev S_3 in the T_0-invariant w = x + 1/x: F4 cost at m = 2 (lead E, 2026-11-02)

Class: component measurement of the `pkm-semaev` oracle (F4 over the bit tower). It claims no change of any ranking and gives no distance to rho. Protocol: PROTOCOL.md, committed (a197a90b8f) before any cell ran. PDF: REPORT.pdf.

Scope of every statement: k = 20 miniature, one curve, m = 2, mode full, t in {4,5,6} (X arm to 7), G in {0,3}, the offsets in the protocol, 100 random targets plus about 20 planted targets per cell, one seed per t. Nothing about m = 3 cost, other windows, other k or curves, the 255-bit setting or a complete DLP.

Answer

Writing the summands in w does **not** lower the solving degree or the F4 cost. In all six primary pairs the solving degree, the polynomial size and the F4 multiply-adds per target are equal (ratio 0.9999 to 1.0019). The reason is algebraic: the polynomial-in-w symmetrisation is the Semaev polynomial of the 2-isogenous curve E/, which has degree 2 in each summand and total degree 4, the same as S_3. The factor 2^(m-1) of STRUCTURE.md is a statement about the presentation `S_3/(z1 z2 z3) = (w1 w2 w3 - D)/2 - 2(w1+w2+w3) - 4A` that needs an extra variable D_i with D_i^2 = w_i^2 - 4d (bilinear in (w_i, D_i)); that variable is not a bit-tower variable and was not built, so the factor is neither confirmed nor refuted for that form. At equal base size the closed base has fewer distinct sums, so the per-relation cost is higher (1.2x to 3.0x). The one cell family where T_0 helps in the generic track (a 2x yield at fixed columns) corresponds to the X arm here, which costs 5.7x to 23x more per relation.

What was built

- `Membership::T0Bits`, `PkmSemaevOracle::new_t0_bits`, `T0Quotient`, `t0_factor_base` in `src/cryptanalysis/ic_framework/pkm_semaev.rs`; `from_spec` accepts `t0_bits=T[,lo=L]`.
- Derivation: with z = x - s (s = x_W(T_0)), d = 3s^2 + a, w = z + d/z; the quotient is y^2 = x^3 + a2 x + b2 at x = w + s, a2 = -3s^2 - 4d, b2 = 2s^3 - 8sd. N(w1,w2,w3) = S_3^{E/}(w1+s, w2+s, w3+s). For d = 1: N (z2 z3)^2 = S_3(z1,z2,z3) S_3(1/z1,z2,z3). In Montgomery terms N = (Q - L)(Q' - L), Q = (W - D)/2, Q' = (W + D)/2, W = w1w2w3, L = 2(w1+w2+w3) + 4A, so N = sum_(i<j) w_i^2 w_j^2 - 4 sum w_i^2 + 16 - L W + L^2, with the leading coefficient in w3^2 equal to (w1 - w2)^2.
- Lift-back charged: target inversion, base map (set-up), two abscissa choices re-added through `lift_abscissae` on the group ledger; counters listed in the protocol.
- `curve25519_family f4cost` accepts `t0-bits[@lo]`; new subcommand `t0size`; `F4COST_PLANTED=n` runs n planted targets after the random ones; JSON gains `base_points`, `lo`, `solver_extra` (matrix size) and `planted_*`.
- Tests: `cargo test --release --lib pkm_semaev`: **test result: ok. 22 passed; 0 failed; 0 ignored; 0 measured; 6086 filtered out** (8 new: quotient is the 2-isogenous curve; S_3-times-its-T_0-image identity; vanishes exactly on valid w-triples, exhaustive over F_229; degree equals the plain degree; F4 = grid = exact decomposable set on random and planted targets for windows (5,0), (6,0), (5,37) with guess 0/1/2; counters; m = 3; a base outside the window is refused). `cargo clippy --release --lib --bin curve25519_family -- -D warnings`: see the session log entry.
- Binary: `/home/user/frozen-bin/curve25519_family_leadE`, sha256 `fa565fd4acf1da0695bdb271f12bbe88cd0064c4bb7be9b10898450a2f098846`.

Table (all 17 cells)

`cols/pts` base columns/points; `deg` max solving degree; `terms` polynomial terms per specialisation; mul-adds per target (F4, elimination only); matrix = max columns x max rows over calls and max nnz; `cov` = `coverage_exact`; `per-rel` = mul-adds per target / cov; `hits/ver` random-target relations / re-summed exactly; `pl` planted found/processed/exact; none capped, none undecided. Cell names: `plain` = `bits` (lo 0), `t0` = `t0-bits` at the protocol's lo, `x` = the secondary arm (cell name `x_tN` has bit-height N).

cell	cols/pts	deg	terms	mul-adds/target	matrix	cov	per-rel	hits/ver	pl	capped	undec
plain_t4_g0	7/14	6	121.0	4046084	243x430, nnz 27671	9e-05	44238198883	0/0	18/18/18	false	0
plain_t4_g3	7/14	4	22.0	88839	31x65, nnz 631	9e-05	971338996	0/0	18/18/18	false	0
plain_t5_g0	11/22	6	256.0	127283713	848x1159, nnz 282072	0.00032	397619167682	0/0	17/17/17	false	0
plain_t5_g3	11/22	5	64.0	3036910	120x186, nnz 6879	0.00032	9486946583	0/0	17/17/17	false	0
plain_t6_g0	25/50	7	484.0	1753646633	3102x2787, nnz 1580643	0.00123	1420269748860	0/0	20/20/20	false	0
plain_t6_g3	25/50	6	154.0	172426718	461x869, nnz 108622	0.00123	139647548220	0/0	20/20/20	false	0
t0_t4_g0	6/12	6	121.0	4046079	243x430, nnz 27671	3e-05	132714438223	0/0	18/18/18	false	0
t0_t4_g3	6/12	4	22.0	88839	31x65, nnz 631	3e-05	2914004853	0/0	18/18/18	false	0
t0_t5_g0	12/24	6	256.0	127283651	848x1159, nnz 282072	0.00026	491176378800	0/0	18/18/18	false	0
t0_t5_g3	12/24	5	64.0	3036908	120x186, nnz 6879	0.00026	11719163058	0/0	18/18/18	false	0
t0_t6_g0	26/52	7	484.0	1757052672	3358x5541, nnz 2458293	0.00072	2452452998101	0/0	20/20/20	false	0
t0_t6_g3	26/52	6	154.0	172425188	461x869, nnz 108639	0.00072	240667042206	0/0	20/20/20	false	0
x_t5_g0	14/28	6	256.0	127284305	848x1159, nnz 282072	0.00027	463891188236	0/0	17/17/17	false	0
x_t5_g3	14/28	5	64.0	3036911	120x186, nnz 6879	0.00027	11068109814	0/0	17/17/17	false	0
x_t6_g0	22/44	7	484.0	1757104723	3358x5540, nnz 2459100	0.00078	2260170696869	0/0	19/19/19	false	0
x_t6_g3	22/44	6	154.0	171995028	461x869, nnz 108616	0.00078	221237879148	0/0	19/19/19	false	0
x_t7_g3	50/100	6	319.0	3459523546	1742x2809, nnz 730851	0.00277	1246977658839	0/0	19/19/19	false	0

Raw: analysis/cells.jsonl (derived), cells/ (per-cell JSON, status with `uptime`, empty stderr). Load average at cell starts: 1.3 to 1.8 (other lead and builds on the box); no cell reached the 600 s cap, so no truncated samples; wall times (0 to 135 s) are not compared.

t0-bits / bits, paired by (t, G), primary base pairs



dashed line = 1; columns 6/7, 12/11, 26/25 at t = 4, 5, 6

ratios

t	G	columns	degree	terms	mul-adds/target	coverage_exact	per-relation
4	0	7 vs 6	6 = 6	equal	1.000	0.333	3.00
4	3	7 vs 6	4 = 4	equal	1.000	0.333	3.00
5	0	11 vs 12	6 = 6	equal	1.000	0.810	1.24
5	3	11 vs 12	5 = 5	equal	1.000	0.810	1.24
6	0	25 vs 26	7 = 7	equal	1.002	0.580	1.73
6	3	25 vs 26	6 = 6	equal	1.000	0.580	1.72

The column mismatch (one column, unavoidable: the closed base has an even number) is inside the coverage ratios: at t = 5 the w base has one column more and covers 0.81 of the plain base, at t = 4 one fewer and 0.33. Matrix size is equal at t = 4, 5 and at G = 3 (nnz at t = 6, G = 3 differs by 0.02 percent); at t = 6, G = 0 the w-form's largest matrix has 5541 rows against 2787 (nnz 2.46M against 1.58M), yet the multiply-adds agree to 0.2 percent; the cause was not investigated.

X arm (one column per class, bit-height t+1), against the plain arm with equal columns

X cell	plain pair	degree X / plain	mul-adds ratio	coverage ratio	per-relation ratio
x_t5_g0	t4 G0	6 / 6	31.5	3.00	10.5
x_t5_g3	t4 G3	5 / 4	34.2	3.00	11.4
x_t6_g0	t5 G0	7 / 6	13.8	2.43	5.7
x_t6_g3	t5 G3	6 / 5	56.6	2.43	23.3
x_t7_g3	t6 G3	6 / 6	20.1	2.25	8.9

Predictions (frozen in PROTOCOL.md)

	claim	outcome
P1 (E)	degree strictly lower in the w-form at G = 0, every t	**falsified**: 6=6, 6=6, 7=7
P2 (D)	degree equal in all six pairs	**held**

	claim	outcome
P3 (D)	polynomial terms equal	**held** (121, 22, 256, 64, 484, 154)
P4 (D)	mul-adds ratio in [0.5, 2]	**held** (0.9999 to 1.0019)
P5 (D)	coverage ratio in [0.3, 0.75]	**falsified at t = 5** (0.81; the w base has 12 columns against 11); 0.33 and 0.58 inside
P6 (D)	per-relation ratio > 1 everywhere, in [1.2, 4] at t = 5, 6	**held** (3.00, 3.00, 1.24, 1.24, 1.73, 1.72; band also held)
P7 (D)	X per-relation ratio > 3 in all five cells	**held** (5.7 to 23.3)
P8	planted all found and exact, hits verified	**held**: planted 18/18, 18/18, 17/17, … all cells found = processed = exact; random hits 0 in every cell, so `hits_verified = hits` is vacuous

Deviations and caveats

1. Base sizes cannot be matched exactly (even closed base versus 7, 11, 25 columns); mismatch one column, reported and priced through each arm's own coverage.
2. The first test run had two failures in my new tests (a planted target equal to T_0, which has no invariant, and an empty base at t = 3). Fixed in the tests only (skip T_0, t = 4); no oracle change.
3. The first frozen binary was rebuilt before any data to add matrix-size fields and the planted option; the cells use the second one (sha above). Source edits to `f4cost` JSON used a map insert because the `json!` macro hit the recursion limit.
4. `agrees_with_grid` gained a sibling `agrees_with_grid_skipping` (the original delegates to it); `montgomery_instance` delegates to `montgomery_instance_from`.
5. Random-target hits are zero in all 17 cells (coverage 3e-5 to 3e-3), so the exactness check on real runs rests on the planted targets and the unit tests.
6. One curve, one seed per t, no spread. Not measured: the (w, D) degree-1 presentation, m = 3 cost in the runs, other windows, G between 0 and 3, wall times.
7. The full t0size scan (162 KB) was not kept; the selection rule and chosen rows are in the protocol.

Coordinator review (main session, 2026-11-02)

- **What was refuted, and what was not.** The w-form tested here writes each summand's w in bits and uses the symmetrised polynomial N of degree 2 per variable. For that form, the expectation of a lower degree at m = 2 is falsified in all six primary pairs: degree, terms and F4 multiply-adds per target are equal to within 0.2%. The STRUCTURE.md factor 2^(m−1) belongs to a different presentation ((w, D) with $D^2 = w^2 - 4d$), which was not built because D is not a bit-tower variable. That presentation is still open, not refuted.
- **Per relation, the w-form is worse here.** At comparable column counts its `coverage_exact` is lower (ratios 0.33, 0.81, 0.58), so per-relation cost is 1.2–3× higher. The X arm, matched on w-class count, is 5.7–23× higher. The T_0-closed base that gave 2× on the generic track gives nothing to the algebraic oracle in these cells.
- **Why the F4 cost is identical.** Both forms put the same bit tower and the same S_3-shaped polynomial (degree 2 per summand, total degree 4) into F4, so F4 sees the same shape. What changes is only which points the bits name.